Estimating Claim Frequencies for Motor TPL Insurance Policies
Warta TUiR Competition

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Abstract - During the summer semester, we participated in a data science competition organized by Warta TUiR. The objective was to predict claim frequency for the next motor third-party liability (TPL; OC in Poland) policy using anonymized market data. Submissions were evaluated exclusively using normalized Gini. Our final ensemble used 565 input features and LightGBM models tuned with Optuna. Performance was assessed across several random seeds; on the organizer's anonymized evaluation data, the submitted model achieved a reported normalized Gini of 25.22%, securing first place in the competition.

I. Introduction

This report documents our work carried out within a student data science competition organized by the Polish insurer Warta, whose objective was to estimate claim frequencies for renewing motor TPL policies. The submitted solution placed first in the competition, with a reported normalized Gini of 25.22% on the organizer's anonymized holdout. The work was conducted as a proof of concept (POC): all analysis and modelling were developed in Jupyter Notebook entirely in Python (pandas, NumPy, scikit-learn, LightGBM, Optuna, and Matplotlib), using anonymized CSV files supplied by the organizer. No production data pipeline or deployment layer was developed.

A. Background Information

Motor insurance is one of the largest segments of the non-life insurance market, while motor third-party liability (TPL) cover is compulsory across the European Union. A central quantity in its pricing is the expected claim frequency of a policy, i.e. the expected number of claims per unit of exposure, where exposure represents the fraction of the policy year during which coverage was active. Accurate estimation of this quantity is important for risk-based pricing, as systematic under- or overpricing can affect both portfolio profitability and fairness across policyholders.

B. State of the Art

Classical actuarial practice addresses this problem with generalized linear models, most commonly Poisson regression with exposure as an offset. As machine learning has developed, pricing teams have increasingly used gradient boosting and tree-based models for tabular data. Deep learning methods are also used, although tree-based methods remain highly competitive for structured tabular datasets.

C. Dataset

The anonymized competition dataset, with identifier fields hashed by the organizer, comprised three CSV tables: a main table of 491,229 observations, each representing a client renewing a motor TPL policy, with information on the vehicle's make and model; and two auxiliary tables containing past policies and past claims. The response variable was the number of claims on the renewed policy, while exposure represented the fraction of the policy year during which coverage was active. Two properties dominated modelling decisions. First, the response variable was extremely imbalanced: 474,397 observations (96.6%) have no claim, 15,564 have one, 1,089 have two, and 179 have three or more (Figure 1). Second, most exposures are close to a full year, while approximately 5% of policies have much shorter exposure. Their observed claim frequencies can therefore be particularly large, making exposure handling essential.

D. Problem Formulation

Submitted predictions were evaluated exclusively with normalized Gini calculated on claim frequencies over an anonymized holdout.

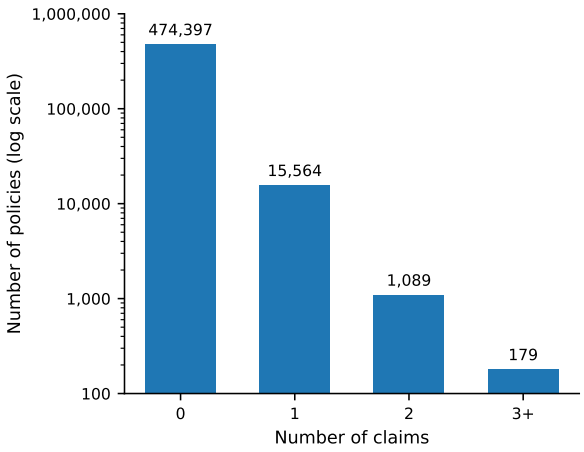


Figure 1: Claim-count distribution on a logarithmic scale, recreated from the counts recorded in the original report draft (not from individual-level data).

Gini is a ranking metric: it depends on the ordering of predictions rather than their absolute values, so calibration was not the competition objective. The strongest signal was expected in clients' own history, which also carried a risk of temporal data leakage. Exposure required particular care because short exposure could itself contain information about the target outcome: a policy may terminate early as a consequence of a claim. Using exposure directly as a predictor therefore introduced target leakage. To avoid this, exposure was excluded from the feature matrix and incorporated only as an offset in the Poisson model. The task was to rank renewing policies in the anonymized scoring set by expected claim frequency, without using information from the target policy period.

II. Feature Engineering

A. Historical Features

Historical policies were aggregated using anonymized identifiers at two levels - per client (UUID) and per client-vehicle pair (UUID-VIN) - mostly by policy type (OC/AC). Each level captured the number and length of past policies, claim/payout/refusal rates, and a snapshot of the most recent policy. Rolling statistics over the last 2/3/5 policies were calculated only per client, and coverage-continuity metrics (gaps between consecutive policies) were calculated for OC policies only.

Claims were aggregated per client over the full history and the last 1, 3, and 5 years, with full-history at-fault and victim breakdowns. Counts were also grouped by policy subtype, claim type, and damage scope, using categories fixed across anonymized training and scoring data. Records with data-quality issues (missing decisions, inconsistent dates, negative payouts) were corrected and flagged rather than removed. Two diagnostic feature types - gaps between claims and registration matches with vehicles from earlier claims - were computed but excluded from the final model.

B. Makes and Models

The raw vehicle make and model entries were cleaned to extract characteristics such as fuel type, body style, and engine capacity. These categories were then used to build smoothed risk indicators through target encoding. To limit the effect of rare vehicles, exposure-weighted smoothing shrank category-level risk estimates with little supporting exposure towards portfolio-wide rates. This helped summarize more than 5,800 vehicle-model categories while reducing overfitting. Crucially, target encoding was performed out of fold within the validation procedure to limit target leakage.

Table 1: Ten highest-ranked features by summed LightGBM gain, with train-scoring KS. Numerical gain values were not retained; the displayed order is the recorded importance ranking.

#	Feature	Description	KS
1	`udzial_polis_bez_szkody_client_oc`	Share of past claim-free OC policies, per client	0.0008
2	`model`	Vehicle model (target-encoded)	0.0196
3	`udzial_polis_bez_szkody_vehicle_oc`	Share of past claim-free OC policies, per vehicle	0.0018
4	`srednia_dlugosc_trwania_client_oc`	Average length of the client's OC policies	0.0016
5	`srednia_dlugosc_okresu_dni_client_oc`	Average continuous-coverage period	0.0023
6	`udzial_polis_ze_szkoda_client_oc`	Share of past OC policies with a claim	0.0012
7	`veh_model_risk`	Vehicle-model risk (target encoding)	0.0623
8	`dni_od_ostatniej_znanej_polisy_client_oc`	Days since the client's last known OC policy	0.0019
9	`veh_model_rate`	Vehicle-model claim rate (target encoding)	0.0526
10	`ostatni_ubezpieczyciel_client_oc`	Insurer of the client's last OC policy (categorical variable)	0.2864

C. Most Important Features

Table 1 presents ten features with the highest summed gain across the 15 base models, together with the Kolmogorov-Smirnov (KS) statistic comparing the anonymized training and scoring distributions. Historical claim-free policy shares and vehicle characteristics were prominent in the gain ranking.

III. Final Model

A. Model Construction

The final model was an ensemble of 15 LightGBM models, obtained from five-fold stratified cross-validation repeated for three different random seeds. Stratification was based on claim-count groups (0, 1, 2, and 3+ claims). Since the response variable represented claim counts, a Poisson objective was used, with the logarithm of exposure incorporated as an offset. Model hyperparameters were selected using Optuna (Table 2), while early stopping was applied independently within each fold to reduce overfitting. To increase prediction stability, outputs from individual models were transformed into percentile ranks and averaged. This aggregation was appropriate for the competition's sole metric, normalized Gini, which evaluates ranking rather than absolute prediction values.

Table 2: Final LightGBM hyperparameters selected using Optuna.

Parameter	Value
Objective	Poisson
Learning rate	0.0236764
Leaves / max. depth	60 / 6
Min. data per leaf	789
Feature / bagging fraction	0.4443 / 0.7790
Bagging frequency	5
L1 / L2 regularization	$2.0723 / 7.63 \times 10^{-6}$
Min. gain to split	0.05952
Max. rounds / early stop	3000 / 100

B. Model Validation

Model performance was additionally evaluated on independent 30% holdout samples using normalized Gini. The model achieved Gini values of 0.2568 and 0.2680 for two recorded splits (Figure 2); for the first split, a 5,000-replicate bootstrap yielded a 95% confidence interval of [0.2382, 0.2753]. Figure 2 shows the bootstrap distributions obtained by repeatedly resampling the 30% holdout sets with replacement and recalculating normalized Gini; the vertical lines denote the medians and the corresponding interval bounds. Potential distribution shifts between anonymized training and scoring data were assessed using the KS statistic for the most important features. The largest KS values occurred mainly for discrete or categorical variables, for which step-shaped empirical distributions warrant cautious interpretation.

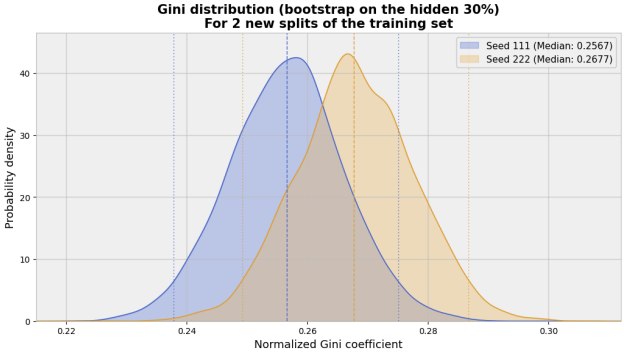


Figure 2: Reconstructed bootstrap distributions of normalized Gini for two independent 30% holdout splits. Dashed lines mark medians; dotted lines mark interval bounds.

IV. Discussion and Limitations

The model was developed for the competition rules and metric. Besides the final specification, we tested smaller feature sets, alternative objectives, and classification-based formulations (see the appendix). The selected model was the best-performing variant under the available validation scheme, but this does not establish production performance. Anonymized training and scoring observations came from the same time period, so temporal and out-of-distribution validation were not available. KS checks suggested similar distributions for most continuous or ordinal features, but they are not a substitute for validation on a later period.

V. Conclusion

The POC combined historical feature engineering, a Poisson LightGBM ensemble, out-of-fold target encoding, early stopping, and rank averaging. The final submission achieved a reported normalized Gini of 25.22% on the organizer's anonymized hidden data and placed first; further work would require temporal validation and production-oriented monitoring.

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Appendix: Finding the Optimal Solution. Two feature sets were tested: an earlier wide set of approximately 455 features and a smaller set of 150-200 features; the wider set achieved around 26.5% normalized Gini compared with approximately 25.9%. After further feature engineering and filtering, the final matrix contained 565 features. We also compared Poisson, binary-classification, and multiclass objectives. Initial classifier scores of 0.296-0.298 were traced to exposure leakage concentrated among short-exposure policies; after removing exposure from the feature set and using it only as an offset, all three objectives converged to around 0.26. Feature capping and monotonicity constraints were also tested but did not produce a consistent improvement, so they were not retained.